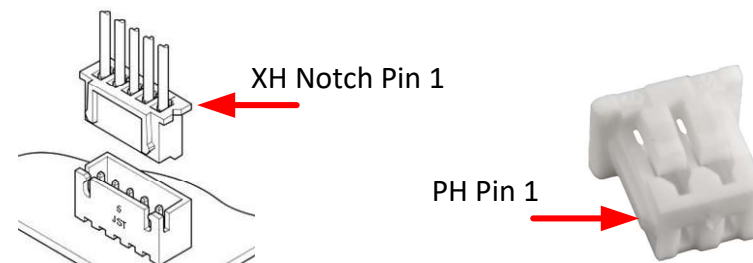


Mellow LLL Buffer For RRF



Items required for the buffer

- 1 x Mellow LLL Buffer
- 2 x JST PHR-2
- 1 x JST XH 3-Way
- 1 x JST XH 4-Way



Note you will need cable and connector for your controll board

Control Board connections required

- 3 inputs,, Filament Detect , Button control Load/unload
- 2 outputs, control buffer load/unload

Example of connections to a Duet 6HC

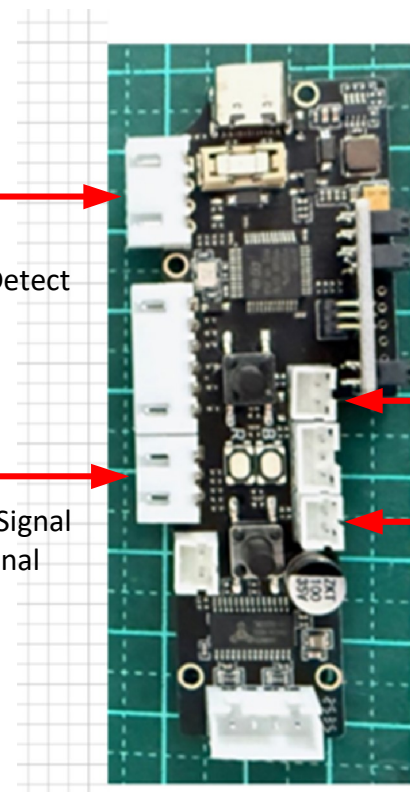
From Buffer	Operation		TO Duet Pin
Retract Control	Retract Filament	control	IO_7_out
Feed Control	Feed Filament	control	IO_8_out
Filament Detect	Check Filament Loaded	signal	IO_8_in
Feed Signal	Button signal Feed	signal	IO_6_in
Retract Signal	Button singnal Retract	signal	IO_7_in

- 1)12/24
- 2)0V
- 3)Filament Detect
- 4)N/C

- 1)0V
- 2)Retract Signal
- 3)Feed Signal

- 1) 0V
- 2) Feed control

- 1) 0V
- 2) Retract control



Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

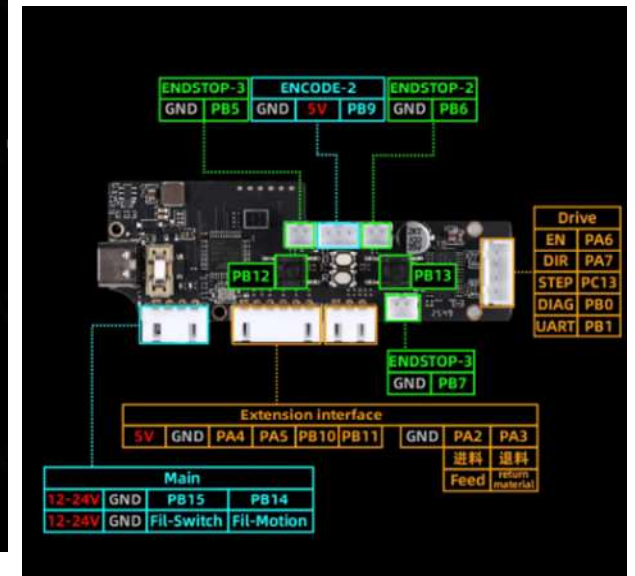
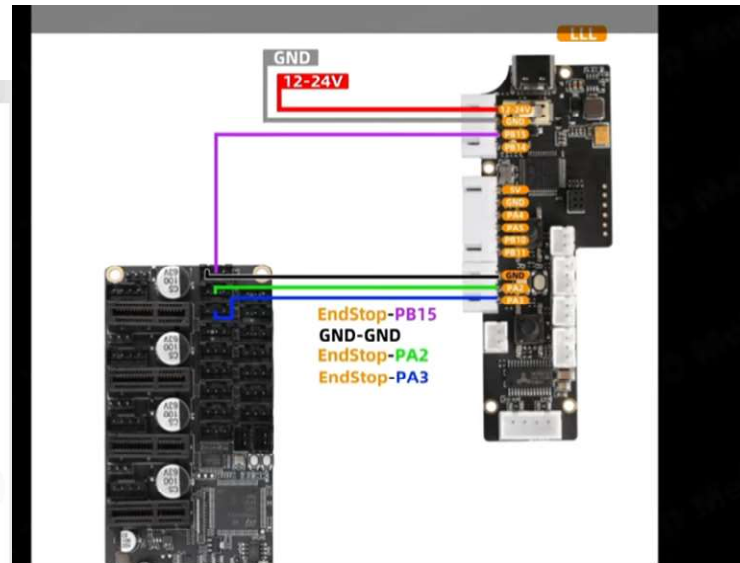
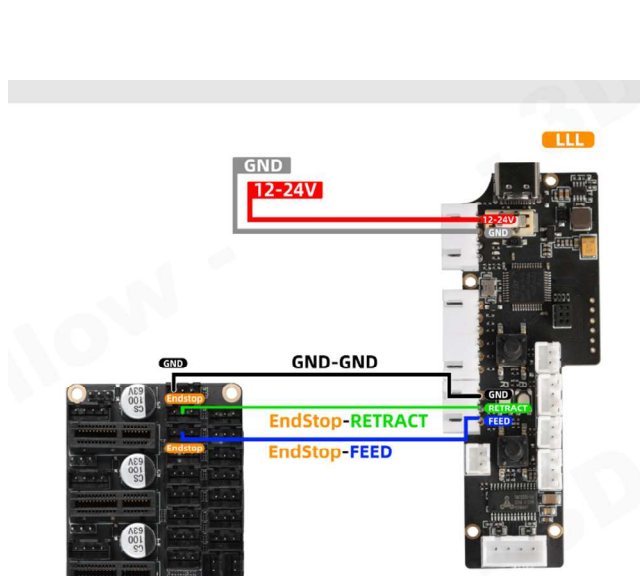
Buffer Pin	Trigger Signal	Action Performed	Duration
Feed Control	Low Level	Buffer continuously feeds filament	Executes while the signal is held low
Retract Control	Low Level	Buffer continuously retracts filament	Executes while the signal is held low

Button	Operation	Signal Output (Buffer Pin)	Signal Type	Duration
Feed Button	Click	FEED pin outputs a high-level pulse	High Level	Automatically returns to low level after 3 seconds
Feed Button	Press & Hold	Continuous feeding	High Level	Until the button is released
Retract Button	Click	RETRACT pin outputs a low-level pulse	Low Level	Automatically returns to high level after 3 seconds
Retract Button	Press & Hold	Continuous retraction	Low Level	Until the button is released

Note: The buffer stops the action when the signal returns to a high level.

How to up date Firmware

<https://mellow.klipper.cn/en/docs/ProductDoc/ExtensionBoard/fly-buffer-plus/update>



Signal Output Description

Button	Operation	Signal Output (Buffer Pin)	Signal Type	Duration
Feed Button (FEED)	Click	FEED pin outputs a high-level pulse	High Level	Automatically returns to low level after 3 seconds
Feed Button (FEED)	Long Press	Continuous feeding	High Level	Until the button is released
Retract Button (RETRACT)	Click	RETRACT pin outputs a low-level pulse	Low Level	Automatically returns to high level after 3 seconds
Retract Button (RETRACT)	Long Press	Continuous retraction	Low Level	Until the button is released

Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

Buffer Pin	Trigger Signal	Action Performed	Duration
PB5	Low Level	Buffer continuously feeds filament	Executes while the signal is held low
PB6	Low Level	Buffer continuously retracts filament	Executes while the signal is held low

Note: The buffer stops the action when the signal returns to a high level.